

1st Examination Date Summer Semester 2026

Lecturer: Benjamin Groß and Prof. Dr. Stephan Huber

Title of examination: Analytical Skills for Business

Study programme: MBA90e, MIMe

Semester: 2

Date: 27.04.2026

The following sections need to be completed legibly by the student:

Name/

Matriculation number: _____

Study programme: _____

DECLARATION OF THE STUDENT ON HIS/HER ABILITY TO TAKE THE EXAMINATION:

The student declares that there are no restrictions regarding his/her mental, emotional and physical state and that he/she is fully capable of taking the examination in its full extent.

Grade: _____

NOTA BENE: PRELIMINARY REMARKS

- Time limit: **90 Minutes**
- Examination aids: **Calculator**
- Mobile phones, Smartwatches and any kind of web-enabled and programmable devices need to be switched off before the beginning of the examination and are not allowed near body and desk!
- The task sheets need to be handed in with your answers.
- If there is more than one lecturer responsible for the examination, questions need to be answered on separate exam sheets.
- All exam sheets that have not been used need to be handed back to the exam supervisor.
- A date for a review of the marked examination paper will be communicated via ILIAS.

This box is for the examiner only:

Question:	1	2	3	4	5	6	Total
Points:	16	17	20	16	15	6	90
Score:							

Analytical Skills for Business (04/2026)

1. Version Control: Collaboration Workflow

- (a) (10 points) Suppose two analysts are jointly working on a forecasting project. Describe a typical collaboration workflow using Git branches and pull requests. In your answer, explain:
- How individual work is organized with branches
 - How changes are reviewed and integrated with pull requests
 - Why this workflow supports effective collaboration
- (b) (6 points) Identify and explain two specific risks that can arise in analytics projects when version control systems are not used. Illustrate each risk with a brief example.

2. Data Types and Suitable Visualizations

- (a) (9 points) Differentiate between categorical data, numerical data, and time series data. For each type:
- Name one appropriate visualization.
 - Explain why this visualization effectively represents the data.
- (b) (8 points) A company wishes to communicate its weekly sales performance to PR managers.
- Propose two distinct visualizations suitable for this purpose.
 - For each visualization, specify a business question it helps to answer and explain your reasoning.

3. Data Quality Management Process

- (a) (10 points) Describe the five-step data cleaning workflow (profiling, validation, cleaning, verification, documentation) and elaborate on the main purpose of each step.
- (b) (10 points) Select one of the following data issues: missing values, inconsistent formats, duplicate records, or outliers. For your chosen issue:
- Provide a brief description of the issue and its impact on data quality.
 - Describe a method for detecting the issue in a dataset.
 - Explain a practical strategy for addressing or resolving the issue.

4. Association and Model Interpretation

- (a) (10 points) In the lecture, we discussed the applications of correlation coefficients in empirical research. Explain what a correlation coefficient measures, and discuss why caution is needed when interpreting correlation coefficients.
- (b) (6 points) Name one key assumption of linear regression. Explain a practical consequence for business analytics if this assumption is violated.

5. No-Code/Low-Code in Analytics Strategy

- (a) (5 points) Select one tool from the course (such as n8n, Tableau, Power BI, KNIME, QGIS, or Streamlit on Snowflake). Describe for the chosen tool a realistic business use case where the tool would be effectively applied.
- (b) (10 points) Explain one situation in which a no-code or low-code solution is preferable to custom programming, and one situation in which custom programming is preferable. For each case, provide reasoning to support your answer.

6. Practical R Programming and Data Wrangling

- (a) (4 points) Write R code to create a tibble named *kpi_data* with two columns:
 - month = Jan, Feb, Mar, Apr
 - revenue = 42, 50, 47, 55

Next, calculate and print:

- mean revenue
 - standard deviation of revenue
- (b) (2 points) Write R code with *ggplot2* to create a line chart of revenue by month.